

Week 6 cross-track handoff

Data Science → Data Analytics

Baseline model error analysis, integrated findings and requested validation

1. Purpose of this handoff

To provide the Data Analytics track with the Data Science findings required to complete the Week 6 cross-track integration and validation activities. It consolidates the baseline model performance, false-positive/false-negative error analysis, comparison with the validated Data Analytics findings, and the specific validation requested from Data Analytics. The intention is not to replace the Data Analytics analysis. Instead, we should use our respective evidence together: Data Analytics validates population level patterns, while Data Science evaluates how those patterns relate to model errors and predictive performance.

2. Week 6 cross-track context

You provided a validated findings package based on the full HealthConnect dataset (n = 5,000). The package was cross-checked against the Data Dictionary and included H1/H2 stability checks for selected findings. The Data Science track used the Week 5 processed dataset for the no-show modelling task. Cancelled appointments were excluded, resulting in 4,737 observations for modelling. Therefore, the Data Analytics descriptive rates and Data Science model error rates should be interpreted as analyses of related but different populations.

3. Data Science baseline model results

The Week 5 Logistic Regression model was re-established in Week 6 using the same modelling features, stratified 80/20 train-test split, and random_state = 42. This provides the reference point for later feature refinement and Random Forest comparison.

Metric	Baseline Result
Accuracy	61.39%
Precision	60.84%
Recall	68.87%
F1-score	64.60%
ROC-AUC	0.6642

4. Baseline confusion matrix and error types

Prediction result	Count	Interpretation
True Negative	248	Attended appointments correctly identified as not being no-shows.
False Positive	215	Attended appointments incorrectly flagged as likely no-shows.

False Negative	151	Actual no-shows incorrectly predicted as attending.
True Positive	334	No-shows correctly identified.

The baseline model identifies a meaningful proportion of no-shows, but it also produces substantial false positives and false negatives. It is therefore a useful reference model, but further development and validation are required before operational use.

5. Data Science error analysis

The Data Science track investigated the distribution of false positives and false negatives across key numerical and categorical variables to determine whether errors were concentrated in particular patient or appointment segments.

5.1 Numerical Error Patterns

Prediction result	Age	Booking lead days	Previous appointments	Previous no-shows	Distance (km)
False negative	48.34	17.58	2.89	0.36	9.08
False positive	48.89	39.11	3.13	0.69	11.49
True negative	50.16	13.29	2.94	0.33	9.02
True positive	47.55	43.39	3.23	0.73	11.02

The largest numerical differences were observed for booking lead time and previous no-show history. False negatives had lower average booking lead time and fewer previous no-shows than true positives. Distance showed some variation, while age and previous appointment counts were comparatively similar.

5.2 Feature-specific error patterns

Feature	Main Data Science observation
Booking lead group	15–30 days and 31–45 days had the highest error rates (~46.5%). 15–30 days had 69 FN; 31–45 days had 100 FP.
Previous no-shows	For patients with 1+ previous no-shows: 183 TP, 45 FN and 105 FP.
Appointment type	Diagnostic Test had the highest error rate (46.1%), followed by Specialist Consultation (42.1%).
Age group	45–54 had the highest error rate (43.2%), followed by 35–44 (42.3%).
Appointment day	Friday had the highest error rate (45.7%), followed by Sunday (43.2%).
Appointment time	Morning had the highest error rate (40.0%); differences were modest.
Distance group	11–15 km had 42.7% error and 16–20 km had 41.2%; 31–50 km was 46.7% but only had 15 observations.

6. Key Data Analytics findings received

The received handoff from your side identified booking lead time and previous no-show history as the strongest predictors of no-show behaviour. Distance was identified as a moderate signal, while appointment type, age group, appointment day and appointment time showed weaker signals.

Finding	Data Analytics evidence	Relevance to Data Science
Booking lead time	27.81% no-show at 0–7 days vs 67.69% at 46–60 days.	Strong agreement with Data Science error patterns.
Previous no-shows	43.51% for 0 prior vs 67.95% for 3 prior.	Supports retaining the feature; rare 4/5 values should be grouped as 3+.
Lead time × previous no-shows	Risk increased as both factors increased; largely additive.	Potential candidate for later feature refinement.
Distance	46.45% at 0–5 km vs 57.76% at 20+ km.	Moderate secondary signal.
Appointment type	Approx. 4.6-point spread.	Lower priority than lead time/history.
Age group	Approx. 5.6-point spread.	Lower priority.
Appointment day	Approx. 3.9-point spread; no weekday/weekend pattern.	Lower priority.
Appointment time	Approx. 1.6-point spread; essentially flat.	Lowest priority.
Reminder channel	Moderate signal; SMS 45.75% no-show rate.	Potential operational lever, but prediction-time availability must be confirmed.

In addition, the handoff received also identified instability in the 8–14 day lead-time band and very small samples for previous no-show levels 4 and 5. They recommended treating previous no-shows as a combined 3+ category for modelling.

7. Cross-track integration: Analytics findings vs Data Science errors

The two tracks show meaningful agreement in several areas. This comparison demonstrates that the Data Analytics findings have been incorporated into the Data Science workflow.

Data Analytics finding	Data Science evidence	Integrated interpretation
Booking lead time is the strongest predictor.	15–30 and 31–45 day groups had the highest error rates; 15–30 had the most FN and 31–45 the most FP.	Prioritise booking lead time during feature refinement and validation.
Previous no-show history is a strong predictor.	Patients with previous no-shows had many correctly detected no-shows, but FP and FN remained.	Retain the feature while investigating its representation.
Distance provides a moderate signal.	Several distance groups showed elevated error rates.	Treat as a secondary candidate.
Appointment type is a weak overall signal.	Diagnostic and Specialist groups had relatively high error rates.	Keep as a possible contributor, but below stronger variables.
Age, day and time are weaker signals.	Their error rates varied, but differences were modest.	Keep as lower-priority candidates.
8–14 day band requires caution.	Data Science identified errors across lead-time groups.	Targeted temporal validation is appropriate before strong conclusions.

8. Important analytical scope difference

The Data Analytics findings use all 5,000 appointments, whereas the Data Science no-show model uses 4,737 records after cancelled appointments were excluded. Consequently, the percentages from the two analyses are not expected to match exactly. This is not a contradiction: Analytics describe the overall dataset, while Data Science evaluates predictive performance on the defined no-show modelling population.

9. Information provided by Data Science to Data Analytics

- Baseline accuracy, precision, recall, F1-score and ROC-AUC.
- Confusion-matrix counts: TN = 248, FP = 215, FN = 151, TP = 334.
- False-positive and false-negative patterns across booking lead-time groups.
- False-positive and false-negative patterns across previous no-show history.
- Error patterns by appointment type, age group, appointment day, appointment time and distance group.
- Identification of booking lead time and previous no-show history as the highest-priority areas.
- Identification of smaller or potentially unstable groups requiring cautious interpretation.

10. Requested validation from Data Analytics

Booking lead-time groups: Validate whether the elevated Data Science error patterns in the 15–30 and 31–45 day groups remain stable across time periods and are supported by adequate sample sizes.

15–30 day false negatives: Investigate whether the concentration of 69 false negatives in this group corresponds to a stable segment-level pattern.

31–45 day false positives: Investigate whether the concentration of 100 false positives in this group is stable and whether the underlying group contains characteristics that may help explain the errors.

Previous no-show history: Review whether the error patterns between patients with zero previous no-shows and those with 1+ previous no-shows remain stable across time.

8–14 day stability: Review this band carefully because the Analytics handoff identified an approximately 11-point H1/H2 difference.

Small sample groups: Confirm whether apparently high error rates in small groups are sufficiently supported by sample size.

Time-split stability: Where possible, apply the same type of H1/H2 or time-split stability assessment used in the Analytics findings.